Mario Becerra, MSc.

Data Scientist / Statistician

Location: Leuven, Belgium

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I am currently doing research in statistics, particularly optimal experimental design for choice models; and have 8 years of experience in Data Science roles. My main interests are data visualization, statistical modeling (particularly Bayesian modeling), experimental design, and predictive analytics.

SKILLS

Statistics and AI

Bayesian statistics, hierarchical/multilevel modeling, exploratory data analysis, Experimental Design, multivariate statistics, choice modeling, A/B testing, machine learning, artificial neural networks

Programming languages and other technologies

Python, R, JMP, Git, C++, SQL, Excel

Other skills

Clear written and verbal communication, leadership, teamwork, adaptability, remote work, time management, consulting

Languages

Spanish: Native English: Native French: Intermediate (B1) Dutch: Intermediate (A2)

EMPLOYMENT

Freelance data scientist and programmer

October 2020 - Present

Small freelance projects that include:
• Data visualization of statistical models and economic data.
• Speed-up of slow **R code** by creating C++ functions on the backend.
• Analysis of Twitter data.
• Choice modeling for market research.

Graduate researcher at KU Leuven, Belgium (Full time)

October 2019 - Present

Research in **statistics**, choice modeling, and **experimental design** in the Biostatistics group at MeBioS.
I created the *opdesmixr* **R package** for computing Bayesian D- and I-optimal designs for choice models involving mixtures of ingredients with the assumption of a multinomial logit model. It uses **C++** under the hood to maximize efficiency. I also published two papers with my supervisor, Peter Goos, using my package. We have one last working paper about designing, performing, and analyzing choice experiments with fruit flies and using a **Bayesian hierarchical logit model** to find their favorite color.
I wrote another paper about modeling the preferences and willingness-to-pay for match tickets of supporters of the Belgian national football teams through statistical analysis of a choice experiment.
I have **presented my research** in eight international conferences and seminars so far. Additionally, I supervised 5 students with their master theses. I provided the students with guidance on their research and code, and gave feedback on their written text.

Data scientist at Banco Azteca, Mexico City (Full time)

September 2018 - September 2019

Data analysis for the marketing and CRM teams of the bank.
I was also in charge of creating and optimizing software for these analyses. Additionally, I managed and helped junior analysts in the area. I also **presented results to non-technical stakeholders**, such as directors and managers of the bank.
Projects included:
• Visualization and analysis of large customer datasets.
• Models for the prediction of future bank transactions.
• Spatial analysis of customer and branch location data.
• Marketing campaign analysis and causal inference.
• Optimized the sampler of a **Bayesian hierarchical model** for BTYD analysis suited for large scale data analysis.
• Creation of marketing campaigns based on customer analysis, particularly:
  - Debit customer segmentation based on regularity of deposits and withdrawals.
  - Identification of highly regular remittance receivers from the US and Mexico.
  - Prediction model to identify remittance customers who are willing to accept a bank loan.

Data scientist at Business Data Evolution, Mexico City (Full time)

September 2017 - September 2018

Developed models both for prediction and for inferential analysis in data science consulting company.
• Analysis of predictive models using Individual Conditional Expectation (ICE) and variable interaction quantification in R.
• Deployed final predictive models in Microsoft Azure VM and Azure Machine Learning.
• Developed Bayesian predictive model for small black and white documents belonging to 4 classes.
• Developed a Bayesian time series demand prediction model for a national grocery store.
• **Met with clients** to understand their problems, and choose the right model for the data and the clients' needs.
• **Presented results** to non-technical audiences.

EMPLOYMENT

Data scientist at CAD Salud, Mexico City (Full time)
May 2016 - June 2017
Built predictive and inferential statistical models for health projects.
- Extracted data from a **centralized database using SQL**.
- Performed **statistical analysis**, exploratory data analysis and developed models in R.
- Created **multivariate predictive model** for potential diabetes patients in public hospitals.
- Generated spatial mathematical model for vaccines demand in public hospitals in Mexico.
- Aided in the creation of multivariate index to rank public hospitals.

Data Scientist at Grupo Salinas, Mexico City (Full time)
October 2015 - May 2016
Designed and built statistical and machine learning models for predictive and inferential analysis in market research.
- Performed statistical analysis, exploratory data analysis and **developed models in R**.
- Created data visualization dashboards using RStudio's Shiny.
- Aided in creation of predictive model of loan origination clients in a Mexican bank.

Data analyst at Computer Research and Analysis Laboratory, ITAM, Mexico City (Part time)
June 2014 - June 2015
Natural Language Processing for automatic news classification for media monitoring company.
- Cleansed news articles data for modeling, **mainly using R**.
- Created different predictive models for automatic news classification in **Python using Scikit Learn**.

Intern at Global Banking and Markets, HSBC, Mexico City (Part time)
December 2013 - June 2014
Financial products pricing, financial products analysis, Excel spreadsheets creation for statistical analysis.

EDUCATION

PhD candidate, KU Leuven, Belgium, (2019 – 2023)
Coursework on **statistics** and **experimental design**. I am mostly dedicated to doing full-time research.

Computer Science MSc, Instituto Tecnológico Autónomo de México (ITAM), Mexico City (2016 - 2018)
Relevant coursework: Machine learning, **Bayesian generalized linear models, Bayesian multilevel modeling**, Big Data, Computer Architecture, Computational Statistics. My thesis was about Bayesian Convolutional Neural Networks.

Applied Mathematics BSc, Instituto Tecnológico Autónomo de México (ITAM), Mexico City (2010 - 2015)
Relevant coursework: Statistical learning, **multivariate statistics**, **Bayesian statistics**, generalized linear models, linear programming, numerical optimization, economics, finance, programming, **databases**. My thesis was about recommender systems.

HOBBIES AND INTERESTS

- **Playing guitar**: I have been playing for 18 years, and I am taking lessons to improve my technique on classical guitar.
- **Running**: at least once a week for a few kilometers each time.
- **Cycling**: I like long-distance road cycling with friends and team members, but I also sometimes do mountain biking.
- **Language learning**: I learned French mostly on my own, and I am currently taking Dutch lessons.
- **Reading**: mostly non-fiction titles, but also classic and modern novels.
- **Traveling**: I like traveling to different countries and cities to experience new cultures.
- **Cooking:** being away from my country made me appreciate and learn the recipes of my culture.
- **Generative art**: I like combining programming and data visualization to create visually appealing images, particularly maps.